

Started on	Thursday, 23 April 2026, 1:08 PM
State	Finished
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Time taken	16 mins 7 secs

Question 1

Complete

Marked out of 1.00

A bioinformatics tool needs to find all occurrences of a DNA pattern within a genome string of length 10^6 . Which data structure gives the most efficient solution?

- ☐ a. Treap on the genome
- ☐ b. Fenwick Tree on the genome
- ☒ c. Trie on the genome
- ☐ d. Suffix Tree on the genome

Question 2

Complete

Marked out of 1.00

For a Segment Tree built on $A = [1, 3, 5, 7, 9, 11]$, what value is stored at the root?

- ☐ a. 18
- ☒ b. 36
- ☐ c. 11
- ☐ d. 21

Question 3

Complete

Marked out of 1.00

For the 'Top K Frequent Words' problem, the PriorityQueue comparator does what?

- ☒ a. Sorts by frequency descending, then alphabetically ascending for ties
- ☐ b. Sorts by frequency ascending, then alphabetically descending for ties
- ☐ c. Sorts by word length descending
- ☐ d. Sorts alphabetically only

Question 4

Complete

Marked out of 1.00

For words = ["abc","bc","ab","qwe"], what does longestWord with all prefixes return?

- ☐ a. abcd
- ☐ b. qwe
- ☐ c. abc
- ☒ d. empty string

Question 5

Complete

Marked out of 1.00

Given FenwickTree input [1, 2, 13, 4, 25, 16, 17, 8] (0-indexed), what is sumRange(2, 6)?

- ☐ a. 80
- ☒ b. 75
- ☐ c. 48
- ☐ d. 86

Question 6

Complete

Marked out of 1.00

Given grid,

```
1 1 0 1 1 0 1 1
1 1 1 1 1 1 1 0
0 1 0 1 0 1 1 1
1 1 1 0 0 1 0 1
1 1 0 0 1 1 1 0
0 1 1 1 1 1 0 1
1 1 1 0 1 0 0 0
1 0 1 1 0 1 0 0
```

number of distinct islands is:

- ☒ a. 1
- ☐ b. 3
- ☐ c. 2
- ☐ d. 4

Question 7

Complete

Marked out of 1.00

Given Treap input [2, 3, 4, 5, 1, 7] with root key=2 (priority=94). After deleting key 2, what becomes the new root key?

- ☐ a. 7
- ☐ b. 1
- ☐ c. 5
- ☒ d. 3

Question 8

Complete

Marked out of 1.00

Graph: 0-1, 1-2, 2-0, 1-3. Bridge is:

- ☐ a. 1-2
- ☐ b. 0-1
- ☒ c. 1-3
- ☐ d. 2-0

Question 9

Complete

Marked out of 1.00

Graph: $1 \rightarrow 2 \rightarrow 3 \rightarrow 1$. Topological sort result?

- ☒ a. Not possible
- ☐ b. Random
- ☐ c. Possible
- ☐ d. Only one order

Question 10

Complete

Marked out of 1.00

In a Fenwick Tree, the expression $i \& (-i)$ is used to:

- ☐ a. Find the parent index
- ☐ b. Reverse the bits of i
- ☒ c. Isolate the last set bit of i
- ☐ d. Count bits in i

Question 11

Complete

Marked out of 1.00

In DSU, Path compression improves:

- ☒ a. Edge count
- ☐ b. Query speed
- ☐ c. Height of tree
- ☐ d. Space

Question 12

Complete

Marked out of 1.00

In DSU, Union by rank helps:

- ☐ a. Increase edges
- ☒ b. Reduce height
- ☐ c. Remove nodes
- ☐ d. Increase height

Question 13

Complete

Marked out of 1.00

In LCA, if node A is ancestor of B:

- ☒ a. LCA is A
- ☐ b. No LCA
- ☐ c. LCA is root
- ☐ d. LCA is B

Question 14

Complete

Marked out of 1.00

In Lexicographically Smallest String Problem, given input: parker morris parser, what is the output?

- ☐ a. makkekr
- ☐ b. makkek
- ☐ c. makkell
- ☒ d. makker

Question 15

Complete

Marked out of 1.00

In lexicographically smallest string problem, union is applied on:

- ☐ a. arrays
- ☐ b. substrings
- ☐ c. indices
- ☒ d. characters

Question 16

Complete

Marked out of 1.00

In topological sort (Kahn's algorithm), we start with:

- ☐ a. Leaf nodes
- ☒ b. Nodes with indegree 0
- ☐ c. Root nodes
- ☐ d. Nodes with max degree

Question 17

Complete

Marked out of 1.00

Initial: Parent = [0,1,2,3], Union(0,1), Union(1,2). Find(2)?

- ☐ a. 2
- ☐ b. 1
- ☐ c. 3
- ☒ d. 0

Question 18

Complete

Marked out of 1.00

Input (Topological Sort): $1 \rightarrow 2$, $1 \rightarrow 3$, $2 \rightarrow 4$, $3 \rightarrow 4$. Valid output:

- ☐ a. 4 3 2 1
- ☐ b. 3 2 1 4
- ☒ c. 1 2 3 4
- ☐ d. 2 1 3 4

Question 19

Complete

Marked out of 1.00

Input: 5 nodes. Edges: 0-1, 1-2, 3-4. Output (connected components)?

- ☒ a. 2
- ☐ b. 4
- ☐ c. 1
- ☐ d. 3

Question 20

Complete

Marked out of 1.00

Lazy Propagation in Segment Trees is used to:

- ☐ a. Balance the tree like an AVL tree
- ☒ b. Defer range updates to avoid $O(N)$ per operation
- ☐ c. Sort the underlying array
- ☐ d. Delete nodes efficiently

Question 21

Complete

Marked out of 1.00

Parallel Courses: Courses: 4, Prereq: $2 \rightarrow 1$, $3 \rightarrow 1$, $1 \rightarrow 4$, $k = 2$. Minimum semesters?

- ☐ a. 4
- ☐ b. 1
- ☒ c. 2
- ☐ d. 3

Question 22

Complete

Marked out of 1.00

Time complexity of bridge-finding algorithm using DFS is:

- ☐ a. $O(M^2)$
- ☐ b. $O(N + M)$
- ☒ c. $O(N \log N)$
- ☐ d. $O(N^2)$

Question 23

Complete

Marked out of 1.00

Topological sorting is applicable only for:

- ☐ a. Cyclic graphs
- ☒ b. DAGs
- ☐ c. Trees
- ☐ d. Undirected graphs

Question 24

Complete

Marked out of 1.00

What defines a connected component in an undirected graph?

- ☒ a. Nodes reachable from each other
- ☐ b. Nodes with no edges
- ☐ c. Nodes with same degree
- ☐ d. Nodes forming cycles

Question 25

Complete

Marked out of 1.00

What is the output of sum(7) on a Fenwick Tree built on values [5, 2, 9, -3, 5, 20, 10, -7, ...] at indices 1–15?

- ☐ a. 25
- ☐ b. 48
- ☐ c. 75
- ☒ d. 36

Question 26

Not answered

Marked out of 1.00

What is the time complexity of both point update and prefix sum query operations in a Fenwick Tree?

- ☐ a. $O(1)$ for both
- ☐ b. $O(N \log N)$ for both
- ☐ c. $O(N)$ for both
- ☐ d. $O(\log N)$ for both

Question 27

Complete

Marked out of 1.00

Which property does a Treap maintain simultaneously?

- ☐ a. BST by key and max-heap by priority
- ☒ b. BST by priority and max-heap by key
- ☐ c. AVL balance and min-heap
- ☐ d. Red-black coloring and max-heap

Question 28

Complete

Marked out of 1.00

Which values are used in articulation point detection?

- ☐ a. degree and weight
- ☐ b. BFS levels
- ☐ c. parent and height
- ☒ d. disc and low

Question 29

Complete

Marked out of 1.00

You need to handle 10^6 queries alternating between point updates and range sum queries on an array of size 10^5 . Which data structure is most appropriate given memory constraints in a contest?

- ☐ a. Fenwick Tree (BIT)
- ☐ b. Plain prefix sum array
- ☐ c. Merge Sort Tree
- ☒ d. Segment Tree

Question 30

Complete

Marked out of 1.00

You're building an autocomplete feature for a search engine. Users type a prefix and you need all words starting with that prefix. Which data structure is best suited?

- ☒ a. Trie
- ☐ b. Binary Search Tree
- ☐ c. Segment Tree
- ☐ d. Hash Map